



Deep Learning Based Air Quality Monitoring: A Downscale Approach

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A.

Introduction

- This project focuses on enhancing N02 spatial resolution by integrating satellite and ground-based data.
- The OMI/Aura satellite provides N02 data at a coarse 25 km x 25 km resolution, which limits city-scale pollution analysis..
- We employ downscaling techniques to refine the resolution to 5 km x 5 km or 5.5 km x 3.5 km for better pollution estimation.
- A deep learning framework is developed, integrating OMI satellite data with CPCB ground measurements to improve spatial accuracy.

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Objective

- Downscale low-resolution N02 satellite data using IDMN and RBF interpolation.
- Integrate CPCB ground data to enhance accuracy and pollutant mapping.
- Provide high-resolution N02 insights for better monitoring and decision-making.

D.

Results

- N02 Hotspots Identified: Major N02 hotspots are concentrated in Northeast India (23-25°N, 92-94°E).
- Enhanced Resolution: Downscaled OMI N02 data from 25 km x 25 km to 5 km x 5 km, enabling finer air quality analysis.
- Seasonal Variation: N02 levels increase in winter (due to temperature inversions) and decrease in summer (due to rainfall- driven dispersion).
- Ground Station Validation: Downscaled OMI data, validated with independent ground stations (unseen in training), shows good agreement using RMSE, MAE, and Bias.
- Challenges in Validation: Discrepancies arise due to unit differences (molecules/cm² vs. µg/m³) and temporal mismatches (daily vs hourly data)
- TROPOMI Comparison: Despite spatial (L2 vs. L3) and temporal differences, downscaled OMI NO₂ aligns reasonably well with TROPOMI data.

C.

Methodology

1. Data Collection & Preprocessing:

- Satellite Data: OMI/Aura NO₂ (25 km × 25 Km), TRO-POMI (5.5 km × 3.5 km).
- Ground Data: CPCB monitoring stations (manual latitude-longitude mapping)
- Preprocessing: Extracted hourly NO₂ data, applied

2. Spatial Interpolation:

- Applied Radial Basis Function (RBF) interpolation to refine NO₂ resolution to 5 km × 5 km.

- A thin plate spline kernel function performed the task of spreading NO₂ concentrations across a 5 km × 5 km resolution grid.

3. Deep Learning-Based Downscaling:

- Used the Improved Deep Mixture Network (IDMN) with a dual encoder to capture both.
- Combined outputs through adaptive fusion (60% satellite, 40% ground) for accurate NO₂ predictions.

4. Model Training & Prediction:

- Trained on combined satellite and ground NO₂ data using Huber loss and adam optimizer (0.001) with early stopping.
- Produced high- resolution NO₂ outputs stored in NetCDF Format.

5. Spatial Visualization of NO₂ Distributions:

- Cartopy was used to map NO₂ distributions, applying a color scale between the 10th and 98th percentiles for clearer visualization across time.

6. Validation & Evaluation:

- Validated Downscaled NO₂ using unseen CPCB stations and TROPOMI data, assessed with RMSE , MAE, Bias metrics.

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Output

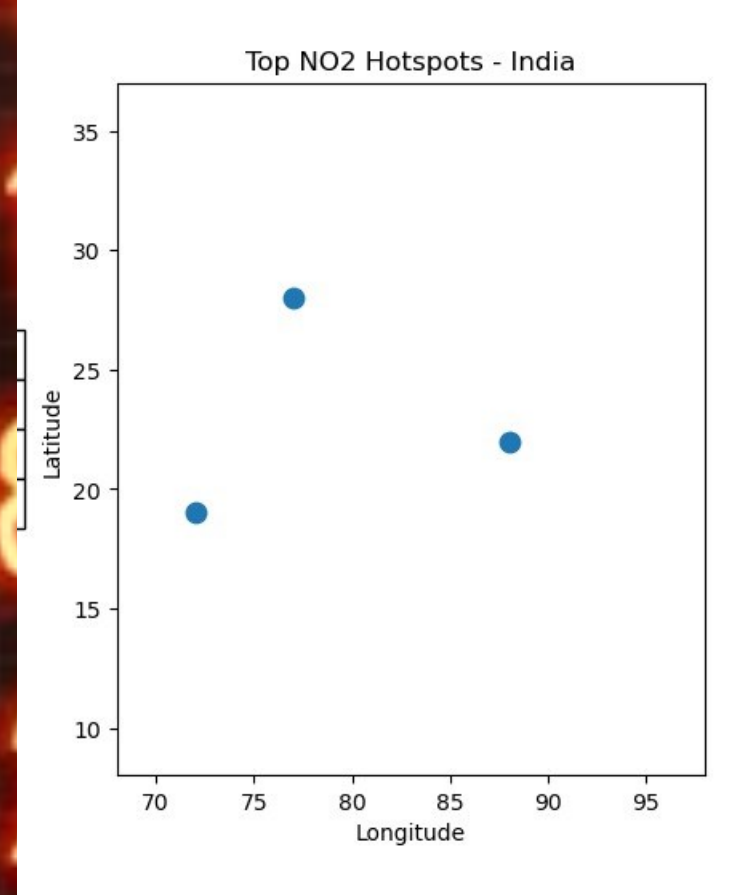
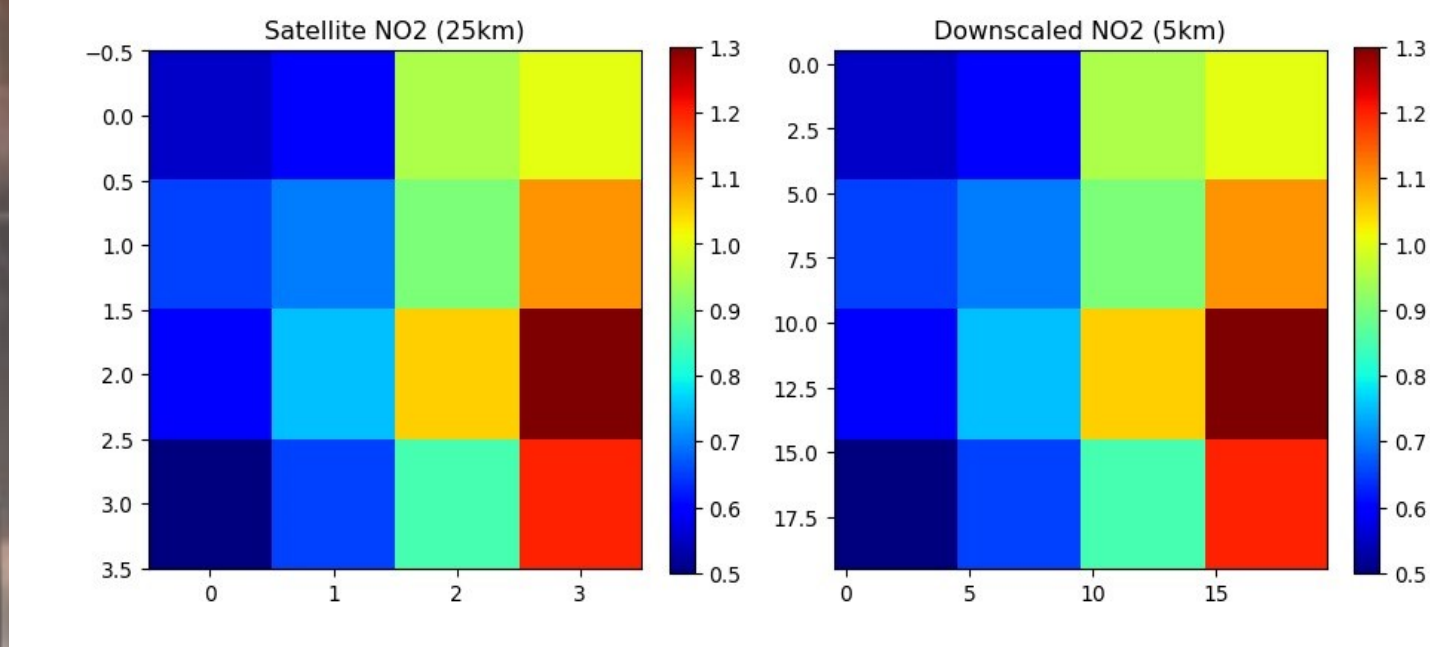
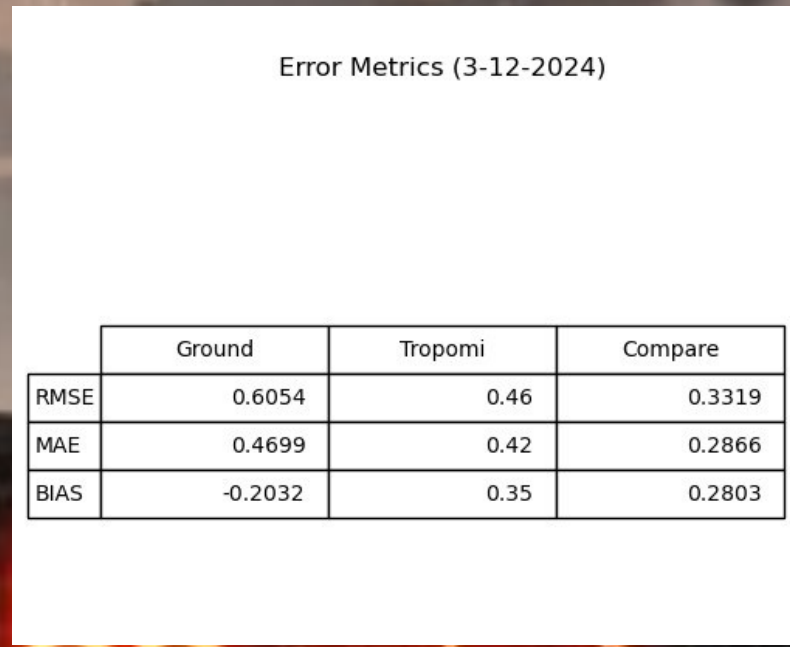
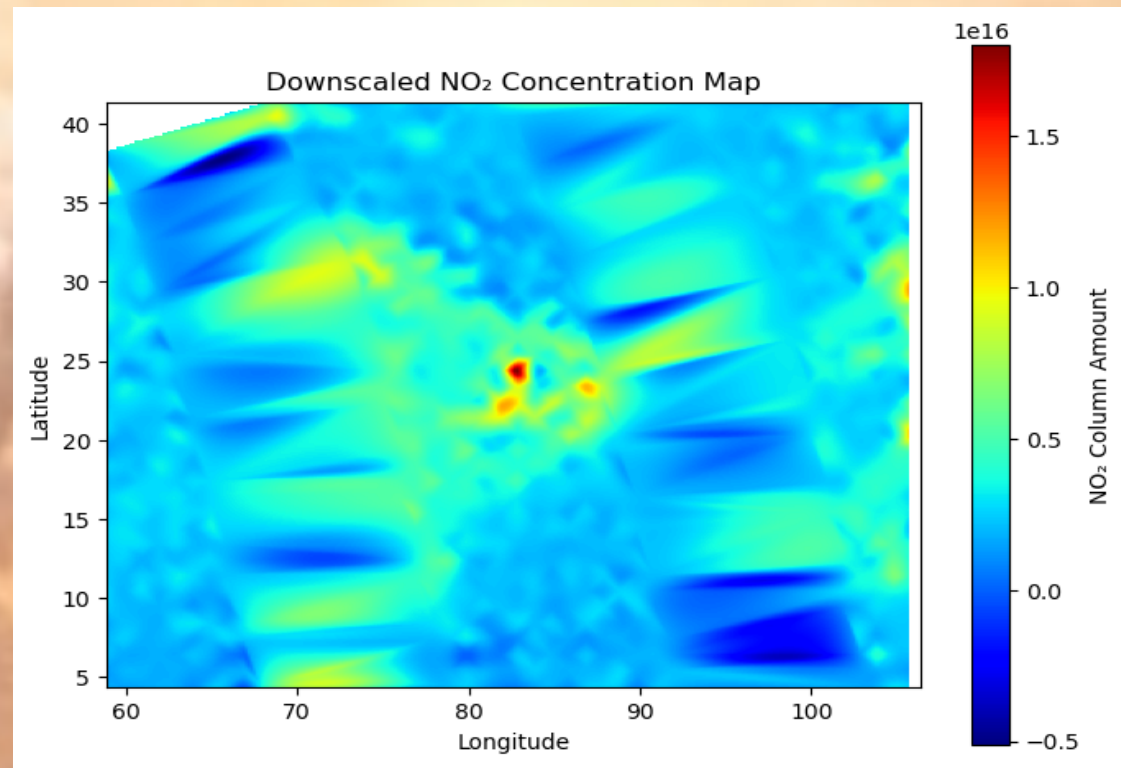
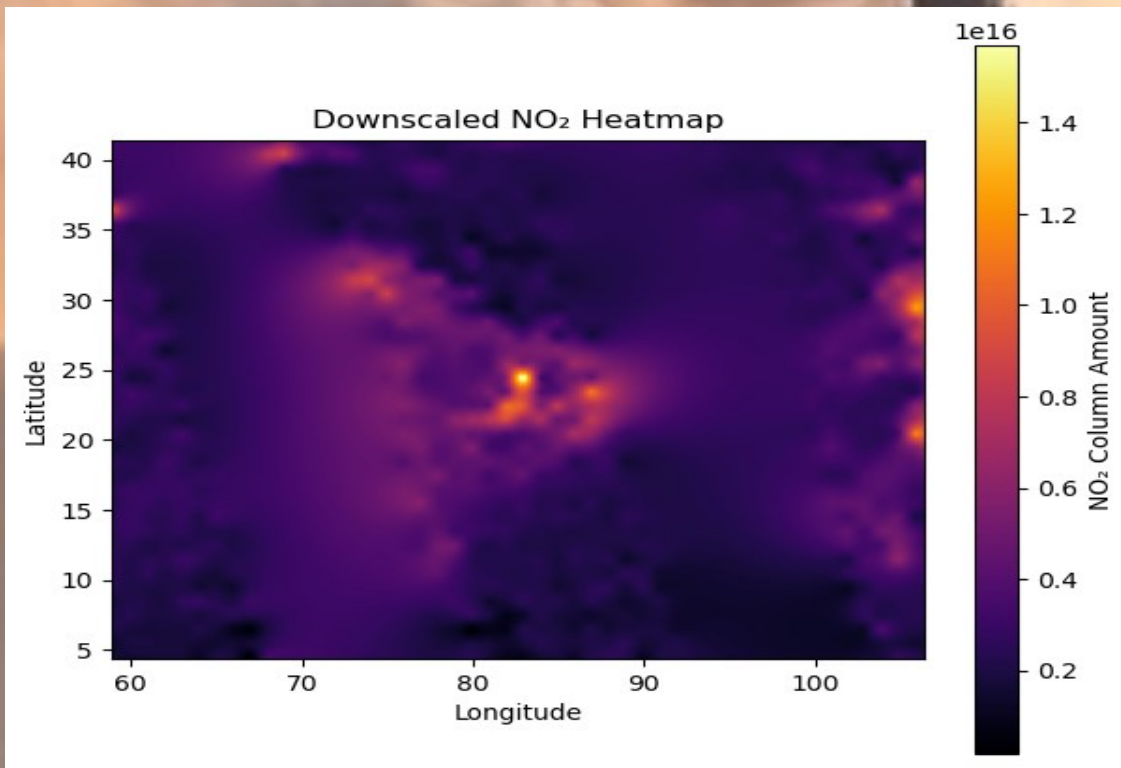


Figure 1: Downscaled NO₂ Heatmap : Displays intensified NO₂ hotspots with improved spatial clarity after downscaling.

Figure 2: Downscaled NO₂ Concentration Map : Shows detailed spatial distribution of NO₂ column amounts across India.

Figure 3: Error Metrics (RMSE, MAE, BIAS) : Validation results indicate reduced errors and good agreement with TROPOMI data.

Figure 4: Satellite NO₂ (25 km Resolution) : Represents coarse-resolution satellite observations with limited spatial detail.

Figure 5: Downscaled NO₂ (5 km Resolution) : Reveals finer-scale pollution patterns and localized hotspots.

Figure 6: Top NO₂ Hotspots – India : Identifies major NO₂ hotspot regions, primarily over North India.

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Conclusion

- Deep learning is used to downscale OMI/Aura NO₂ 25 km × 25 Km to 5 km × 5 km, improving accuracy.
- Validation shows strong agreement with TROPOMI NO₂ data and low error metrics.
- Discrepancies occur due to cloud interface and temporal mismatches.
- The study identifies NO₂ hotspots and seasonal variations linked to Industries, vehicles and weather.
- Enhanced resolution supports better air quality assessments for policymakers.

References

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- [2] Hao Zhang Yongqian Liu, Jie Yan, Member, IEEE, Shuang Han, Li Li, and Quan Long (2020), **Improved Deep Mixture Density Network for Regional Wind Power Probabilistic Forecasting**; Article in Power Systems, IEEE Transactions on February 2020; DOI: 10.1109/TPWRS.2020.2971607